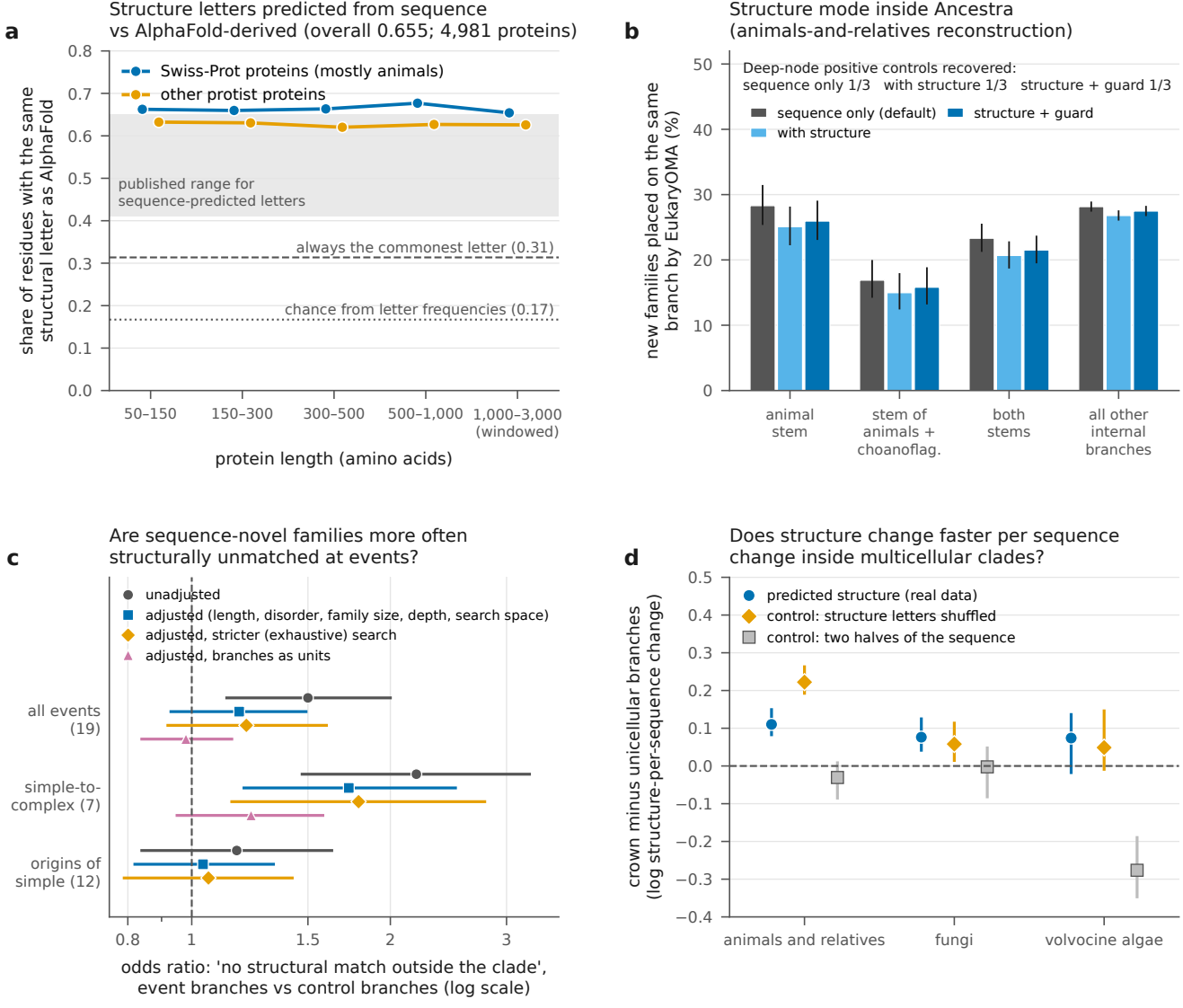
a
Structure letters predicted from sequence
vs AlphaFold-derived (overall 0.655; 4,981 proteins)
Swiss-Prot proteins (mostly animals)
other protist proteins
published range for
sequence-predicted letters
always the commonest letter (0.31)
chance from letter frequencies (0.17)
share of residues with the same
structural letter as AlphaFold
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
50–150
150–300
300–500
500–1,000
1,000–3,000
(windowed)
protein length (amino acids)
b
Structure mode inside Ancestra
(animals-and-relatives reconstruction)
Deep-node positive controls recovered:
sequence only 1/3 with structure 1/3 structure + guard 1/3
sequence only (default)
structure + guard
with structure
new families placed on the same
branch by EukaryOMA (%)
0
10
20
30
40
50
animal
stem
stem of
animals +
choanoflag.
both
stems
all other
internal
branches
c
Are sequence-novel families more often
structurally unmatched at events?
unadjusted
adjusted (length, disorder, family size, depth, search space)
adjusted, stricter (exhaustive) search
adjusted, branches as units
all events
(19)
simple-to-
complex (7)
origins of
simple (12)
0.8
1
1.5
2
3
odds ratio: 'no structural match outside the clade',
event branches vs control branches (log scale)
d
Does structure change faster per sequence
change inside multicellular clades?
predicted structure (real data)
control: structure letters shuffled
control: two halves of the sequence
crown minus unicellular branches
(log structure-per-sequence change)
−0.4
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4
0.5
animals and relatives
fungi
volvocine algae